Lab 8 Rubric: Buoyancy

Out of 100 points. This is the rubric on the LB8-M6 assignment in Brightspace. Lab 8 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains, so read them before you write rather than after.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence, capped at 3 points for the report. A fault already scored under a criterion is not deducted again as a formatting occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase "human error" earns no credit anywhere in the report. A percent error under 5% against 1.000 g/cm^3 is a good result for this apparatus. Treating Routes A and B as independent confirmations of each other is a misreading of your own data — they share the displaced-volume measurement — and is marked as such in Discussion.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab title and date, all present and correctly formatted. No template placeholders.	**2** One item missing or misformatted.	**2** Template placeholders left in the heading block.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer. States what was measured, gives the density of water reported and by which route, its percent error against 1.000 g/cm^3, and the identity of the metal. No methods.	**6** All content present but runs long or includes method detail.	**4** Reports a density but does not say which route produced it.	**2** Describes the lab but reports no numbers.	**0** No abstract.
Introduction	8	**8** Derives ρ_water = ρ_metal(1 − T_water/T_air) line by line from the free-body diagram and Archimedes' principle, showing the volume cancel, with every symbol defined and the full-submersion assumption stated. Ends with a testable hypothesis.	**7** Derivation present but the cancellation or the assumption is thin.	**5** The relation is quoted rather than derived.	**3** Archimedes' principle stated with no derivation.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Apparatus in prose with the precision of each instrument; a labelled sketch showing the cylinder in air AND in water with forces marked in both.	**7** Sketch present with forces; instrument precision not given.	**5** Only one of the two situations sketched, or forces unmarked.	**3** Bare equipment list, or no sketch.	**0** No materials or setup description.
Experimental: Procedure	8	**8** Written in own words as prose, reproducible. Says how the water level was read and to what precision, and how full submersion without contact was ensured.	**7** Reproducible and in own words; the precision statement is absent.	**5** Reproducible but silent on the submersion condition.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.
Results: Data Tables	12	**12** Trial table with all three dips and the averages, plus a results table giving the density of water by all three routes with percent errors. Captions above, units in header rows, significant figures that respect the volume reading.	**10** Both tables present; minor issues with captions, units or significant figures.	**7** Only one route reported, or the trial averages missing.	**4** Fewer than three trials tabulated.	**0** No data tables.
Results: Graphs	8	**8** No graph is required for this lab; this criterion covers the presentation of the derived quantities instead. Full credit for a clear results table or bar comparison of the three routes against the accepted value, correctly labelled with units.	**7** Comparison present but labelling incomplete.	**5** The three routes are reported but never set side by side against 1.000 g/cm^3.	**3** Results scattered through the prose with no organised presentation.	**0** Nothing presented.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Results: Sample Calculations	13	**13** One worked sample of each type with units at every step: displaced volume, density of metal, buoyant force, and each of the three routes. The newtons-to-g/cm^3 conversion in Route A shown explicitly.	**11** All types present with minor unit or rounding slips.	**8** Three or four of the six types worked.	**4** Some types missing, or the Route A unit conversion skipped.	**0** No sample calculations.
Discussion	15	**15** Opens with the reported density and the route it came from. Explains that Routes A and B share the displaced-volume measurement and are therefore not independent, while Route C is. Uses the identity of the metal as an external check and names the measurement responsible for the discrepancy.	**12** Most of the above; the independence argument is present but thin.	**9** Compares the routes but treats their agreement as corroboration without noticing the shared input.	**5** Restates the results without interpreting them.	**0** No discussion.
Additional Questions	8	**8** Three of the five chosen and named, each answered fully in a dedicated section, in paragraphs with working shown and not restating the question.	**6** Three answered but one thinly, or the three chosen are not named.	**4** Answered inside another section, or only two answered.	**2** Answered as a bare list, or fewer than two answered.	**0** Not answered.
Conclusion	5	**5** Three to four sentences: numerical result, definitive statement about Archimedes' principle, one concrete improvement.	**4** Present but missing the numbers or the verdict.	**3** Present with numbers but no verdict.	**2** Vague, or a repeat of the discussion.	**0** No conclusion.
References	5	**5** At least one source beyond the textbook plus a density reference; Note 1 and the handout count. Author, title and date, consistently formatted.	**4** Multiple references but formatting incomplete.	**3** Textbook and handout only, properly formatted.	**2** Handout only.	**0** No references.
Total	100					

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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